Frequency–Phase Aware Mamba Fusion for Multiscale Spatial Representation in Remote Sensing Image Interpretation

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***Abstract*—Remote sensing images (RSIs) exhibit complex spatial patterns due to diverse object scales, orientations, and distributions. These characteristics pose significant challenges for accurate interpretation. While existing methods focus on spatial feature extraction or rotational constraints, they often overlook the intrinsic relationship between frequency–phase information and spatial structures, which is critical for precise localization and classification. In this article, we propose a generic and flexible feature fusion framework, called phase-aware Mamba fusion mechanism (PMFM), which jointly captures frequency–phase patterns and global spatial dependencies. PMFM consists of two core modules, i.e., frequency–phase position encoding (PhasePE), which enhances positional encoding by embedding frequency–phase cues, and cross-attention–Mamba progressive fusion (CaMPF), which integrates global–local information across multiscale feature maps using cross-attention and vision Mamba. Furthermore, we design a PMFM-based feature pyramid architecture (PMFM-FPN) via a cross-scale concatenation strategy, enabling seamless integration into various detection and segmentation frameworks. Extensive experiments on five public RSI benchmarks (DOTA, DIOR-R, SISP, ISPRS Potsdam, and Vaihingen) across three major tasks, i.e., horizontal detection, oriented detection, and semantic segmentation, demonstrate that PMFM consistently improves performance and achieves competitive accuracy and robustness. In addition, an auxiliary experiment on ExDark further explores the behavior of phase-aware representations under general low-light visual conditions.**

***Index Terms*—Feature fusion, frequency–phase modeling, object detection, remote sensing image (RSI) interpretation, semantic segmentation, vision Mamba.**

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I. Introduction

REMOTE sensing images (RSIs) interpretation is of great importance in both computer vision and Earth observation, playing an essential role in applications such as marine monitoring [1] and airspace planning [2]. Different from natural images, RSIs capture geographic scenes from a unique bird-eye perspective, resulting in objects that appear in arbitrary orientations with significant variations in scale, angle, and spatial location. Such distinctive imaging characteristics produce complex spatial structures and diverse distributions of target bounding boxes, posing unique and significant challenges for effective scene-specific understanding.

Currently, convolutional neural networks (CNNs) [3] and vision transformers (ViTs) [4] are the primary frameworks employed in mainstream RSI interpretation tasks. CNN-based methods [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] have improved spatial representation by refining feature extraction and incorporating rotational constraints. Nevertheless, these methods inherently rely on local representations [16] and tend to exhibit high-frequency bias [17], restricting their ability to handle the complex spatial structures typical of RSIs. In contrast, Transformer-based approaches [18], [19], [20], [21], [22] effectively address rotation invariance through positional encoding and global self-attention mechanisms, leading to superior performance in capturing long-range spatial dependencies.

However, these approaches primarily model low-frequency, semantic information, lacking the ability to adequately represent critical high-frequency details, such as textures and edges [23]. Furthermore, conventional positional encoding strategies (e.g., sinusoidal position encoding (SinPE) [24], RelativePE [25], AbsolutePE [4]) primarily focus on relative spatial positions, disregarding the intrinsic characteristics and content of the image signals themselves [26]. As illustrated in Fig. 1(b), the commonly used SinPE uniformly encodes spatial relationships without considering intrinsic differences in target-background image content, leading to imbalanced spatial modeling. This limitation may reduce recognition performance in complex RSI scenarios.

Recently, several approaches [27], [28], [29], [30] have introduced frequency-domain analysis into natural image interpretation, demonstrating notable improvements by effectively quantifying and distinguishing high-value feature channels.

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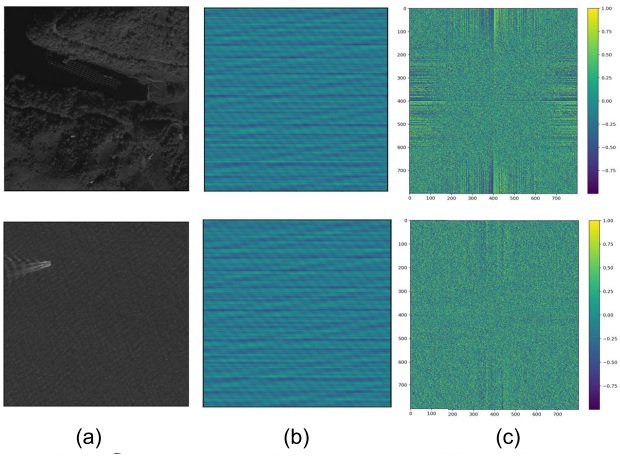


Fig. 1. Encoding visualization of different images. (a) Original image. (b) SinPE. (c) PhasePE. The visualization shows that SinPE does not consider the natural characteristics of the image signal, and the same coding scheme is obtained for images with significantly different target-background context. PhasePE introduces frequency–phase information of the image at the position encoding stage as a distinct supervisory signal, promoting the differentiation and representation of image content within the spatial information modeling process.

For example, FreqFusion [27] and FADC [29] adopt dynamic feature extraction and adaptive sampling mechanisms based on frequency information, emphasizing high-frequency features while modulating the ratio between high- and low-frequency components. Despite their effectiveness for natural images, these methods exhibit reduced adaptability to RSI interpretation tasks, due to intrinsic domain differences such as resolution variations, unique imaging geometries, and complex backgrounds.

To address these domain-specific challenges, frequency-domain analysis has also been explored in recent RSI interpretation studies [31], [32], [33], [34]. HLRSCompNet [31] and MFGCN-CME [32] introduce specialized modules to simultaneously represent high-frequency and low-frequency features. LFA [33] and LSCNet [34] utilize frequency-based modules to enhance global discriminative feature representations. Furthermore, methods specifically tailored to RSI characteristics have also been proposed. For example, FDINet [35] leverages wavelet interactions to effectively analyze spatial–temporal correlations. SF-MSFormer [16] integrates spatial–frequency multiscale features using parallel Transformer branches and texture-enhanced encoders, while TSFNet [36] exploits frequency amplitude information to reconstruct complex textures in RSI super-resolution tasks. Nevertheless, existing methods generally neglect the critical interplay between frequency-domain features and spatial positional modeling. This oversight leads to inaccurate representation of targets with arbitrary spatial distributions and orientations, resulting in misalignment between object regions and background contexts, thus limiting their robustness in analyzing complex RSI scenes.

On the other hand, some studies have explored the combination of Transformers with feature pyramid networks (FPNs) [37], employing a cross-attention mechanism to capture global information from multiscale features for enhanced prediction [19]. However, existing cross-attention structures often fail to adequately explore the correlation between global context and spatial details across multiple feature scales, limiting model performance in complex scenarios. Although recent work has achieved more robust global feature representations by improving self-attention methods, effectively modeling interactions among cross-layer and multiresolution feature maps remains challenging [38], [39], [40], [41]. Furthermore, these methods often struggle to balance accurate representation of target features against background interference, which may limit their applicability to RSI tasks with complex backgrounds.

To address these limitations, in this article, we propose a novel phase-aware Mamba fusion mechanism (PMFM) for effective integration of frequency-domain and spatial-domain information in RSI interpretation. Our PMFM introduces two key modules: frequency–phase position encoding (PhasePE) and cross-attention–Mamba progressive fusion (CaMPF). Specifically, the PhasePE module embeds frequency–phase features into positional encoding by characterizing the spatial characteristics (i.e., initial positions, displacement trajectories, and intensity variation patterns [42]) of frequency components. This dynamic positional encoding paradigm effectively balances spatial structure representation, addressing limitations inherent in traditional encoding methods when dealing with complex RSI scenes. Moreover, the CaMPF module progressively integrates cross-attention mechanisms with vision Mamba, establishing a robust fusion architecture that deeply explores multiscale global and spatial dependencies. The vision Mamba mechanism dynamically aggregates contextual information across spatial positions and orientations, effectively alleviating correlation estimation losses caused by noncausal visual data. Concurrently, the cross-attention component progressively refines global semantic representations from shallow to deep layers, facilitating accurate modeling of cross-scale spatial dependencies. In addition, the PMFM mechanism is seamlessly integrated into various FPNs via a cross-scale concatenation strategy, demonstrating strong generalization and robustness across different architectures and interpretation tasks. As an example of implementation, we construct the PMFM-based FPN (PMFM-FPN) to illustrate the integration paradigm within the representative feature fusion framework. In conclusion, the optimized PMFM-FPNs achieve the multiscale spatial representation and fusion capability of global–contextual and frequency–spatial features, improving object detection and segmentation performance across different settings, especially for arbitrarily oriented objects. Extensive experiments conducted on multiple RSI benchmark datasets, including DOTAv1.0 [43], DIOR-R [44], SISP [45], ISPRS Potsdam [46], and Vaihingen [47], with the additional low-light benchmark ExDark [48], demonstrate the effectiveness and generalization capability in addressing complex spatial scenarios across various object detection and image segmentation tasks.

The main contributions of this article are summarized as follows.

1) We propose a novel multiscale feature fusion mechanism called PMFM, which effectively integrates global–contextual and frequency–spatial information. The proposed PMFM is a generic and flexible approach

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that can be flexibly integrated into various FPNs to form a PMFM-FPN, substantially enhancing multiscale feature representation, particularly beneficial for complex RSI scenarios.

2) We introduce an interpretable frequency-aware positional encoding module, named PhasePE, which explicitly models frequency–phase characteristics alongside spatial position information. By modeling spatial information (initial positions, displacement trajectories, and intensity variations), PhasePE effectively resolves spatial modeling imbalances in conventional positional encoding for RSI tasks.
3) We design a CaMPF module, effectively integrating cross-attention with the vision Mamba architecture. CaMPF progressively aggregates multiscale global context and spatial dependencies across feature maps, improving robustness and representation capability for RSI interpretation.
4) Extensive experiments are conducted on five RSI benchmarks, together with auxiliary low-light validation, across three downstream tasks, i.e., object detection with horizontal and oriented bounding boxes, and semantic segmentation. The results demonstrate the effectiveness, robustness, and generalization capability of the proposed PMFM-FPN across diverse RSI scenarios.

The remainder of the article is organized as follows. Section II reviews and analyzes previous related work. Section III introduces the design idea and implementation details of the proposed methods in detail. Section IV provides our experimental results on various datasets. Section V discusses the effectiveness and performance. Section VI concludes the article.

II. Related Work

In this section, we discuss three types of subjects: frequency domain information analysis in image processing, multiscale feature fusion in RSIs, and vision Mamba in image processing.

A. Frequency Domain Information Analysis in Image Processing

Frequency-domain information reveals intensity variation patterns critical to image understanding. In general computer vision tasks, recent studies such as FreqFusion [27], FADC [29], and FSNet [28] enhance high-frequency representation via adaptive sampling or attention on frequency-decomposed features. FSEL [30] further explores frequency–spatial modeling with entangled Transformer blocks. However, these methods focus primarily on frequency magnitude and overlook phase information, which encodes crucial spatial structure cues. In addition, due to differences in resolution and layout, they are not directly applicable to RSIs. To address domain-specific challenges, several methods have incorporated frequency modeling into RSI tasks. LFA [33], LSCNet [34], and FDINet [35] enhance feature learning through spectral priors and wavelet decomposition. TSFNet [36] and HLRSCompNet [31] improve structural preservation and multifrequency representation. However, most of these methods still neglect frequency–phase modeling and its interaction with spatial position. To bridge this gap, we propose PhasePE to explicitly integrate frequency–phase information into positional encoding, thereby enhancing frequency–spatial representation and improving spatial modeling in complex RSI scenes.

B. Multiscale Feature Fusion in RSIs

FPNs [37] have been widely adopted in RSI interpretation due to their ability to integrate multiscale features for improved object prediction. Early variants focused on optimizing convolutional structures and path aggregation, such as PAFPN [49], BiFPN [50], and NAS-FPN [51], which employed multibranch aggregation to enhance feature fusion. Subsequently, more refined designs emerged to address misalignment in multiscale feature maps. To this end, some approaches introduced attention mechanisms [38] and feature alignment modules [52], [53], [54], [55] to improve spatial consistency during fusion. With the rise of Transformers in vision tasks, FPNs have evolved to incorporate Transformer-based modules for enhanced long-range dependency modeling. For instance, FPT [38] introduces cross-scale self-attention to capture nonlocal interactions, while TransMIN [39] and PSTNet [40] utilize cross-view and progressive sampling Transformers for spatial-channel and pixel-wise refinement. Recent studies such as FSCNet [41], EIA-PVT [18], and FPNFormer [19] further exploit frequency–spatial attention and hierarchical cross-attention to improve scene understanding across scales.

Despite these advances, two aspects remain insufficiently explored. First, the intrinsic content of image signals, such as frequency–phase cues, can provide useful guidance for spatial representation. Second, the interaction between global context and spatial structure across scales is still not fully modeled. To address these challenges, we propose PMFM-FPN, which incorporates frequency–phase information into positional encoding and combines Transformer and Mamba-based architectures for multiscale feature fusion, enabling more effective joint modeling of local structures and global semantics in RSI tasks.

C. Vision Mamba in Image Processing

Vision Mamba [57], known for its linear complexity and global receptive field, has shown strong potential in long-sequence modeling [58] and global context extraction [59], [60]. Its recent extensions to RSI tasks focus on improving encoding strategies or integrating Mamba into other architectures [61], [62], [63], [64], [65], [66]. For instance, RSMamba [63] introduces dynamic multipath activation to improve non-causal modeling. Multimodal variants like S2CrossMamba [64] and RSMamba-Diffusion [65] fuse spectral and spatial information to enhance geometric consistency, while GraphMamba [66] and MGMF [67] explore graph-based or cross-modal fusion. Recently, several advanced Mamba-based detectors have been introduced for RSIs, such as LS-MambaNet [68], which employs large kernels for long-range modeling, OriMamba [69], which adopts a hybrid pyramid design for local–global integration, and DFLAN [70],

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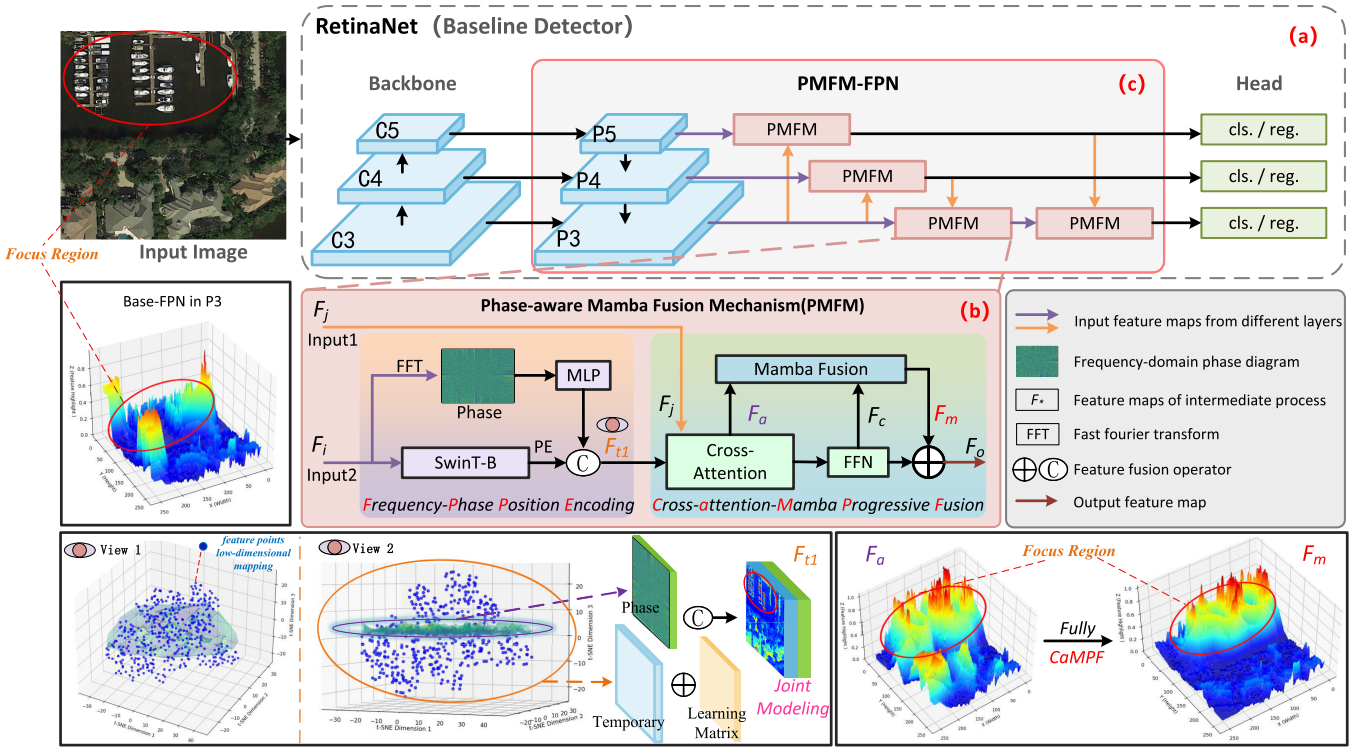


Fig. 2. Overview of the proposed PMFM with the deployment on a simple baseline detector, RetinaNet [56]. The proposed PMFM consists of two components: PhasePE and CaMPF. Here, (a)–(c) correspond to the subsections of Section III, respectively. For the input image (top left) and feature map visualization, "focus region" indicates the target of interest (e.g., ship, harbor). The visualized feature maps illustrate how activations evolve across processing stages. "Base-FPN in P3" (left) denotes the FPN fused representation from P_3 level, corresponding to the blue P_3 layer in (c). "Views 1 and 2" illustrate the PhasePE enhanced embedding process of the feature maps F_{t1}, showing how phase-informed positional cues affect the representation. "F_a, F_m" (lower right) correspond to the refined feature maps produced by the cross-attention stage (F_a) and by the full CaMPF pipeline (F_m), respectively, demonstrating progressive concentration of activations on true target regions. The PMFM-FPN is a generic feature fusion network that seamlessly integrates PMFM to enhance multiscale feature representation through a cross-scale concatenation strategy.

which focuses on contextual modeling for small objects. Although effective in global sequence modeling, these methods pay limited attention to hierarchical feature alignment and spatial–structural representation capabilities that are essential for accurate target localization in complex backgrounds of RSIs.

Different from these approaches, our CaMPF establishes a progressive fusion strategy that explicitly models the frequency–spatial collaborative relationship across scales. Cross-attention serves as the shallow fusion stage, providing frequency-aware spatial cues and cross-level alignment. Mamba Fusion then functions as the deep fusion stage, refining and propagating these cues through long-range state-space scanning to capture deeper structural dependencies. This complementary interaction effectively compensates for the weaker local alignment capability of standalone Mamba modules, enabling more robust multiscale spatial representation.

III. Methodology

In this section, we introduce the proposed PMFM for multiscale spatial representation in RSI interpretation. We first present the architecture of the baseline detector. Then, we detail how our PMFM comprising PhasePE and CaMPF is integrated into the FPN, forming the PMFM-based FPN.

A. Architecture of the Baseline Detector

To validate the effectiveness of the proposed mechanism, we select a lightweight anchor-based detector, RetinaNet [56], as our baseline. Our fusion mechanism optimizes the feature fusion architecture of RetinaNet by jointly modeling global contextual dependencies and frequency–spatial information across multiresolution feature maps. The overall processing flow is illustrated in Fig. 2.

As shown in Fig. 2(a), RetinaNet follows the standard detection pipeline: a backbone network, a feature fusion network, and detection heads. Let $G \in \mathbb{R}^{H\times W\times 3}$ denote an input image of size $H \times W$. After feature extraction, the backbone outputs a set of hierarchical feature maps $F_1, F_2, \ldots, F_n$, where $F_i \in \mathbb{R}^{H_i\times W_i\times D_i}$ represents the ith layer with resolution (H_i, W_i) and channel dimension D_i. Traditional FPN receives higher level feature maps C_3, C_4, and C_5 to generate fused multiscale feature maps P_3, P_4, and P_5. These are subsequently fed into detection heads to predict classification $\hat{Cls}_i$ and bounding box regression $\hat{Reg}_i$ outputs.

The proposed PMFM is integrated into the feature fusion stage to enhance cross-scale interaction. While retaining the original FPN to preserve early cross-level feature semantics, PMFM further processes multilevel features P_i, P_j through the proposed PhasePE and CaMPF modules to model spatial–global interactions. This forms a hierarchical fusion architecture, referred to as PMFM-FPN, as shown in Fig. 2(c),

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which improves multiscale representation and feature fusion quality.

To facilitate a clear analysis of our method, PMFM is first integrated into the lightweight RetinaNet [56], serving as a minimal baseline. To further evaluate its generality, in our study, PMFM is also applied to advanced detectors, including Swin Transformer [71], RTMDet [72], and S^2A-Net [5], and segmentation models such as SegFormer [73], MaskFormer [74], and Mask2Former [75], demonstrating robust performance across diverse RSI tasks.

Section III-B introduces the proposed PMFM, including its two core modules, PhasePE and CaMPF, and then Section III-C describes how it is integrated into a unified feature fusion framework, namely, PMFM-FPN, as illustrated in Fig. 2.

B. Phase-Aware Mamba Fusion Mechanism

Fig. 2(b) shows our PMFM, comprising PhasePE and CaMPF. PhasePE embeds frequency–phase signals into positional encoding to form explicit correlations between phase variations and spatial coordinates, thereby enhancing the representation of complex RSI scenes. Meanwhile, CaMPF integrates cross-attention blocks with vision Mamba in a multistage manner, progressively capturing global context from shallow to deep layers. This hierarchical cross-scale fusion helps refine local spatial details and long-range semantic representations.

Below, we detail each module and then describe how they integrate into the overall PMFM pipeline.

1) Frequency–Phase Position Encoding: RSIs are characterized by diverse spatial distributions and orientations of objects, often leading to challenges in accurate position modeling when using only standard positional encodings such as SinPE [24] or RelativePE [25]. To address this, we propose PhasePE, which explicitly incorporates frequency–phase cues into positional encoding.

a) Frequency–phase extraction: Let F_i denote the input feature map at the ith level from the baseline neck, with a size of $H \times W \times D$. We perform a 2-D fast Fourier transform (FFT) to obtain the complex-valued frequency representation, as follows:

$$F_i^{freq}(u,v,c) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} F_i(h,w,c)\, e^{-\mathrm{j}2\pi\left(\frac{hu}{H}+\frac{wv}{W}\right)} \tag{1}$$

where $F_i^{freq}(u,v,c)$ denotes the complex frequency representation at frequency coordinates (u,v) for channel c of the ith feature map, decomposing into a real part R and an imaginary part I. We then compute the phase map as

$$F_i^{Phase}(u,v,c) = arctan\left(\frac{I(u,v,c)}{R(u,v,c)}\right). \tag{2}$$

b) Phase-guided positional encoding: The extracted phase map $F_i^{Phase}(u,v,c)$ is fed into a multilayer perceptron (MLP) for dimensional alignment and representation enhancement, allowing the model to capture complex patterns in the frequency–phase representation.

Then, we concatenate it with the original feature F_i after a Swin Transformer block (i.e., $SwinT$), which retains a RelativePE-style learnable weight matrix L_i, thereby effectively integrating frequency-domain and spatial-domain information in RSI interpretation. The enhanced feature representation F_{t1} is expressed as

$$F_{t1} = Concat\left(MLP\left(F_i^{Phase}\right),\quad (SwinT(F_i) + L_i)\right) \tag{3}$$

where the feature map $SwinT(F_i)$ and the weight matrix L_i are combined through element-wise addition with broadcasting. Specifically, $SwinT(F_i)$ has dimensions (B,C,H,W), while L_i has dimensions $(1,C,H,W)$. Therefore, L_i is broadcast along the batch dimension and added to $SwinT(F_i)$ at each spatial position. This operation injects relative positional information into the transformed spatial features before the subsequent concatenation step, thereby enhancing the structural representation.

By embedding frequency–phase signals directly into the positional encoding, PhasePE emphasizes crucial edge, texture, and geometrical cues, which can be beneficial for robust orientation estimation and complex background suppression in RSIs. In Fig. 2 (Views 1 and 2), the t-distributed stochastic neighbor embedding (t-SNE) visualization suggests that PhasePE produces more clearly separated feature distributions, resulting in more discriminative feature clusters.

2) Cross-Attention-Mamba Progressive Fusion: To further integrate the encoded frequency–phase feature with global semantic context in multilevel feature maps, we propose a progressive two-stage fusion strategy, i.e., CaMPF, which models the frequency–spatial collaborative relationship through the joint use of cross-attention and vision Mamba. As shown in Fig. 3, within the CaMPF module, the distinction between "shallow" and "deep" is not intended to describe computational cost, but rather the functional order of information processing under a "focus-then-refine" paradigm. The 'shallow' cross-attention stage applies multihead self-attention over cross-level features from the FPN, enabling alignment of spatial cues and extraction of frequency-informed interactions before deeper processing. This stage focuses on local-to-regional refinement and emphasizes regions of interest in the early fusion phase. The "deep" Mamba Fusion stage is constructed using residual-stacked vision Mamba blocks. It performs dual-branch state-space scanning to capture long-range dependencies and enhance global structural coherence. This deeper sequential reasoning consolidates and propagates the cross-attention-enhanced cues across scales, yielding more robust spatial representation for foreground–background separation.

a) Shallow cross-attention: During the shallow modeling process, the cross-attention module computes explicit correlations between global and spatial features across hierarchical feature maps using the standard Q,K,V formulation of multihead attention.

Consider two feature maps, F_{t1} from PhasePE and another multiscale feature F_j. The Q,K,V for multihead self-attention can be calculated as

$$Q = F_{t1}W_q + b_q \tag{4}$$

$$K = F_jW_k + b_k \tag{5}$$

$$V = F_jW_v \tag{6}$$

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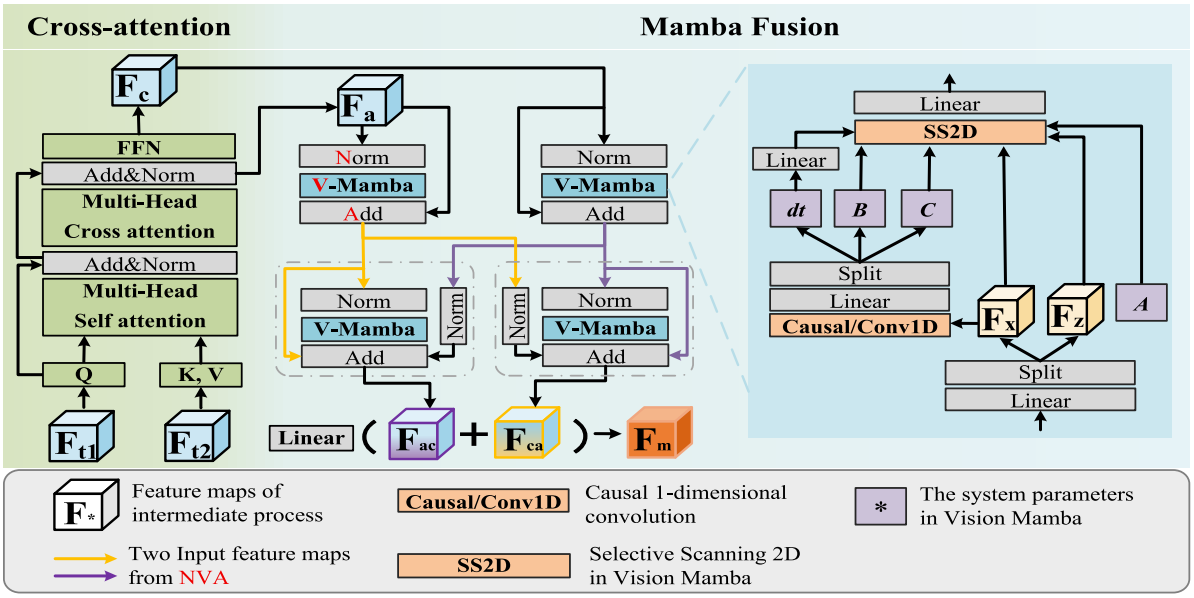


Fig. 3. Overall structure of the proposed CaMPF component, which is constructed from cross-attention and Mamba Fusion. The cross-attention module uses multihead self-attention to extract shallow global features and facilitate spatial interaction across cross-level feature maps. The Mamba fusion module is built on residual-stacked vision Mamba blocks and implements a dual-branch interaction strategy. By leveraging the SS2D mechanism, it performs deep contextual information extraction over feature maps with different resolutions.

where W_q, W_k, and W_v are the coefficient matrices of query Q, key K, and value V in self-attention, respectively. b_q, b_k are the bias matrices. Then, F_a is obtained after multihead self-attention as

$$F_a = MHA(Q, K, V). \tag{7}$$

A feed-forward network (FFN) then refines F_a to enhance local feature representation

$$F_c = FFN(F_a). \tag{8}$$

This shallow cross-attention explicitly aligns and augments feature interaction between layers of potentially different resolutions.

b) Deep Mamba fusion: To complement the global and local information modeled by cross-attention, we feed F_a, F_c into the Mamba Fusion module to extract long-range contextual features. The computation flow is shown in Fig. 3

$$F_a^* = NVA(F_a) \tag{9}$$

where NVA denotes stacked residual blocks consisting of normalization, vision Mamba, and residual addition. F_a^* and F_c^* are computed by the same symmetric computation to obtain deep semantic features.

As a computing core, vision Mamba reshapes the input features into a 2-D patch sequence, denoted as $X_t \in \mathbb{R}^{L\times D}$. Calculated by selective scanning 2-D (SS2D), the intermediate implicit states $h_t \in \mathbb{R}^{L\times D}$ and the system parameters (the projection feature maps $D = \{F_x, F_z\}$, the state transition matrix $A \in \mathbb{R}^{N\times N}$, the time-step d_t and control state parameters $\{B, C\}$) are updated, to obtain the output $Y_t \in \mathbb{R}^{L\times D}$. The computation is described below

$$F_x, F_z = LS(F_a) \tag{10}$$

$$d_t, B, C = LS(CausalConv(F_x)) \tag{11}$$

where LS denotes the channel splitting operation after linear mapping, and $CausalConv(\cdot)$ is used to capture causal dependencies in the state-space transition process. Furthermore, an interactive fusion process is designed to extract deep semantic feature relationships, aiding the model in understanding spatial information and cross-scale features. The calculation process is described below

$$F_{ac} = NVA(F_a^*) + Norm(F_c^*) \tag{12}$$

$$F_{ca} = NVA(F_c^*) + Norm(F_a^*) \tag{13}$$

where $Norm(\cdot)$ denotes normalization operation. Finally, the linear layer is used to integrate features. The output feature map is obtained as $F_m = Linear(F_{ac} + F_{ca})$. The final output feature map F_o from the proposed PMFM is obtained by fusing F_m and F_c.

C. PMFM-Based FPN

Building on PhasePE and CaMPF, PMFM establishes a frequency–spatial collaborative modeling paradigm, simultaneously leveraging frequency–phase information and global spatial interactions, further enhancing multiscale feature representation to make it particularly beneficial for complex RSI scenarios. Notably, PMFM can be seamlessly integrated into various FPN architectures via a cross-scale concatenation strategy.

For illustration, we construct an example PMFM-based FPN (i.e., PMFM-FPN) by inserting PMFM into neighboring layers of a standard FPN, as shown in Fig. 2(c). Specifically, we introduce additional skip pathways to augment multiscale information exchange from bottom to top. For instance, consider the following cross-scale fusion process among layers P_3 to P_5:

$$D_5 = PMFM(P_5, P_3) \tag{14}$$

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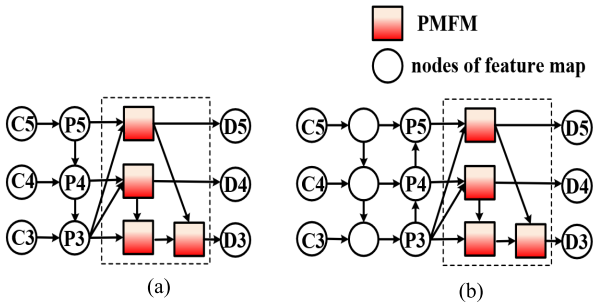


Fig. 4. Combination of the proposed PMFM with typical FPN structures, i.e., (a) PMFM-FPN and (b) PMFM-PAFPN.

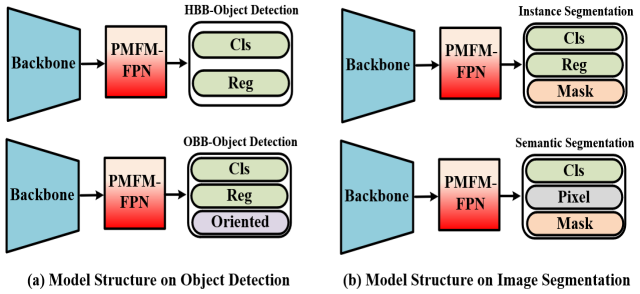


Fig. 5. Combination of the proposed PMFM-FPN with four typical downstream tasks, including (a) horizontal and oriented object detection and (b) instance segmentation and semantic segmentation.

$$D_4 = PMFM(P_4, P_3) \tag{15}$$

$$D_3 = PMFM(PMFM(D_4, P_3), D_5) \tag{16}$$

where $PMFM(\cdot)$ denotes our PMFM. The fused representations $\{D_3, D_4, D_5\}$ capture complementary features across different resolutions and orientations. To improve compatibility, PMFM adopts a feature alignment strategy for feature maps with different spatial resolutions and channel dimensions. For resolution mismatch, the internal cross-attention module within PMFM is designed to accommodate different spatial resolutions through adaptive up- or down-sampling, enabling long-range interactions across scales. For channel mismatch, we adopt 1×1 point-wise convolutions on all backbone feature maps (e.g., C_3:256, C_4:512, C_5:1024) to project them into a unified channel dimension (256), consistent with standard FPN processing. These alignment steps ensure PMFM can seamlessly handle interactions across adjacent and nonadjacent feature levels.

Beyond a standard FPN, PMFM can readily be combined with other neck structures, such as PAFPN [49] and FPN-Former [19], yielding flexible and effective multiscale feature fusion. Fig. 4 provides an abstract illustration of how PMFM integrates into two typical feature fusion frameworks. The circular nodes represent the processing flow within the feature maps. The dashed box outlines the suggested stacking example in this section, with the red rectangles indicating the proposed PMFM, while the solid lines illustrate the cross-scale concatenation strategy.

Meanwhile, Fig. 5 further extends the PMFM-FPN concept to various downstream tasks (e.g., object detection and image segmentation), showing that a straightforward block-based integration scheme can align the backbone output channels with the prediction heads.

By capturing multiscale global semantics and frequency–phase information alongside cross-layer attention, PMFM-FPN improves detection and segmentation performance for objects with diverse sizes and orientations in complex RSI scenes.

IV. Experiments and Results

We conduct extensive experiments on several publicly available and challenging RSI datasets to evaluate the effectiveness of the proposed method. The following subsections present details on the experimental setup, effectiveness validation, ablation studies, generalization analysis, and visualization results.

A. Experimental Setup

1) Datasets: To evaluate the effectiveness and robustness of the proposed mechanism, we conduct experiments across three representative RSI interpretation tasks, including horizontal object detection, oriented object detection, and semantic segmentation. Five publicly available and challenging datasets are used as benchmarks.

a) Oriented object detection: We use two popular large-scale datasets, i.e., DIOR-R [44] and DOTAv1.0 [43]. DIOR-R contains 23 463 images with 192 518 oriented bounding box annotations across 20 classes. All images are used at their original resolution of 800×800 pixels for both training and inference. DOTAv1.0 contains 2806 aerial images with 188 282 annotated instances across 15 classes. During training, images are cropped to 600×600 pixel patches with a stride of 450, and resized to 800×800. A multiscale strategy with 1024×1024 crops and scales of $[0.5, 1.0, 1.5]$ is employed to enhance training diversity. The test set is evaluated using the official evaluation server.

b) Horizontal object detection: We adopt the SISP dataset [45], which contains 10 000 panchromatic satellite images and 56 693 annotated ship instances across four fine-grained categories, i.e., civil ship (CS), platform (PL), auxiliary ship (AS), and military ship (MS). Unlike RGB images, panchromatic images consist of grayscale channels only, posing additional challenges for object detection. We follow the official train and test splits, and apply random flipping for data augmentation.

c) Semantic segmentation: We use two popular large semantic segmentation datasets, i.e., ISPRS Potsdam [46] and Vaihingen [47]. Potsdam includes 38 high-resolution aerial images (6000×6000 pixels) across six classes (i.e., impervious surface, low vegetation, tree, car, building, clutter) and four spectral bands (i.e., RGB and near-infrared). Vaihingen contains 33 fine-resolution images (2500×2000 pixels) with three spectral bands (near-infrared, red, and green) and the same classes. Both datasets follow the official training, validation, and test splits. For training, only RGB bands are used. The images are cropped into 1024×1024 patches, and standard data augmentation (e.g., random flipping and scaling) is applied.

d) Auxiliary low-light validation: Beyond mainstream interpretation tasks of RSIs, we further introduce ExDark [48] as an auxiliary low-light visual benchmark to examine the

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TABLE I

COMPARATIVE EXPERIMENTS TO VALIDATE THE SOTA OF THE METHODS IN THIS ARTICLE ON THE DIOR-R DATASET, WHERE "†" INDICATES THE REIMPLEMENTATION RESULTS. "*" MEANS THE PROPOSED PMFM-FPN IS USED

Methods	Venue	APL	APO	BF	BC	BR	CH	ESA	ETS	DAM	GF	GTF	HA	OP	SH	STA	ST	TC	TS	VH	WM	mAP_{50}(%)
FCOS [77] †	ICCV'19	62.3	29.9	74.2	80.9	28.7	72.1	70.4	55.4	25.9	65.8	75.8	37.3	45.1	80.6	74.4	69.0	81.2	43.9	43.5	62.3	58.9
RoI-Transformer [10]	CVPR'19	63.3	37.9	71.8	87.5	40.7	72.6	68.1	78.7	26.9	69.0	82.7	47.7	55.6	81.2	78.2	70.3	81.6	54.9	43.3	65.5	63.9
Gliding Vertex [78]	TPAMI'20	65.4	28.9	75.0	81.3	33.9	74.3	64.7	70.7	19.6	72.3	78.7	37.2	49.6	80.2	69.3	61.1	81.5	44.8	47.7	65.0	60.1
R^3Det [9]	AAAI'21	62.6	43.4	71.7	81.5	36.5	72.6	79.5	64.4	27.0	77.4	77.2	40.5	53.3	79.7	69.2	61.1	81.5	52.2	43.6	64.1	61.9
FASSROD [79]	TGRS'22	61.5	36.3	72.2	87.9	42.1	71.0	75.7	70.2	31.5	74.2	81.8	45.6	55.0	82.0	73.4	70.9	81.5	**61.0**	51.0	68.9	64.8
TIOD-Det [80]	ISPRS'23	68.7	28.6	76.7	84.8	39.3	72.4	72.7	63.9	20.5	75.2	77.4	40.6	47.5	82.6	72.6	70.3	81.9	47.9	52.1	64.2	62.0
FCOSF [21]	TGRS'23	61.1	47.5	74.1	81.2	35.4	73.1	78.2	63.0	29.4	74.8	80.1	40.7	54.2	80.3	62.0	67.4	81.3	57.7	42.4	64.8	62.4
CGCDet [14]	TNNLS'23	68.5	38.3	**79.1**	86.2	39.0	73.5	74.7	66.0	26.8	67.5	**84.5**	48.0	56.1	81.3	**79.3**	72.2	88.6	50.7	51.7	65.8	64.9
QPDet [11]	TGRS'23	63.2	41.4	72.0	**88.6**	41.2	72.6	69.0	**78.9**	28.8	70.1	83.0	47.8	55.5	81.2	72.2	62.7	**89.1**	58.1	43.4	65.4	64.2
HRDet [12]	TCSVT'24	71.2	28.7	78.6	81.5	36.3	72.6	67.2	76.8	23.9	69.4	82.1	43.8	52.3	81.2	70.7	62.7	82.0	54.1	43.1	64.8	62.1
RPG-S^2A-Net [15]	TGRS'24	71.4	51.7	76.6	87.8	**46.8**	73.2	84.1	74.2	31.5	72.0	82.2	46.6	**59.7**	85.9	60.7	70.0	88.6	59.3	53.5	65.3	67.1
ARS-DETR [22]	TGRS'24	68.0	**54.2**	74.4	81.7	41.1	75.7	81.9	73.1	**34.9**	76.1	78.6	36.3	55.4	84.6	70.1	72.2	81.1	61.5	50.6	**70.3**	66.1
DFDet [13]	TGRS'24	61.9	38.8	77.4	81.4	34.1	75.0	76.1	62.3	26.3	75.6	79.6	38.3	52.8	80.4	73.1	68.3	81.4	52.2	44.1	63.4	62.1
OrientedFormer [20]	TGRS'24	65.7	48.7	78.8	87.2	41.9	76.3	81.4	72.1	34.4	75.3	79.8	45.2	56.1	88.7	67.6	72.7	87.3	60.3	**56.5**	69.6	67.3
SOOD [81]	TGRS'24	62.6	46.8	72.0	81.3	41.4	72.6	78.6	68.9	27.5	**78.6**	81.4	46.9	56.3	80.6	73.7	68.4	81.4	56.9	47.3	65.5	64.4
RetinaNet [56]	ICCV'17	59.2	27.3	76.2	80.3	29.3	72.7	76.6	60.2	20.9	78.5	74.9	40.9	43.8	78.8	74.4	65.2	80.0	52.1	42.3	58.3	59.6
RetinaNet*(**ours**)		70.7	26.9	75.5	79.1	31.9	72.5	78.0	62.7	29.2	78.4	77.4	42.3	44.9	79.0	73.1	66.3	80.3	48.5	43.1	62.3	**61.1(+1.5)**
RTMDet [72] †	arXiv'22	70.0	50.6	75.2	81.5	42.1	77.0	85.9	70.5	33.1	77.1	80.3	**49.1**	57.7	89.2	66.9	77.2	81.6	57.3	51.6	65.2	66.9
RTMDet*(**ours**)		**78.5**	46.5	75.4	86.6	43.3	**77.7**	**86.4**	71.3	32.8	75.3	80.2	48.2	58.6	**89.5**	64.4	**78.7**	87.8	55.6	53.2	65.9	**67.8(+0.9)**

behavior of the proposed phase-aware representation under degraded illumination conditions. ExDark is a real-world low-light image dataset designed for visual recognition under challenging illumination conditions. It contains 7363 images captured in ten different low-light environments, ranging from extremely dark scenes to twilight, with both image-level annotations and object-level bounding boxes for 12 object categories. These results complement the remote sensing experiments by providing auxiliary observations on the behavior of phase-aware representations under general low-light visual conditions, without being used as direct validation of remote sensing-specific illumination robustness.

2) Evaluation Metrics: To evaluate the performance of the detectors, we select the widely used metrics that allow for fair comparisons between the performance of our method and the state-of-the-art (SOTA) methods. In the object detection task, the mean average precision (mAP) with intersection over union (IoU) is adopted to evaluate the accuracy of detectors. Both mAP_{50} and $mAP_{50:95}$ are selected as the benchmark metrics to evaluate the overall performance of the model, while the average precision (AP) evaluates performance for each category in the dataset. For the semantic segmentation task, mean $F1$-score (mF1) and mean IoU (mIoU) are used to evaluate segmentation accuracy. In particular, we compute the mIoU of the five foreground classes for the ISPRS Potsdam and Vaihingen datasets. The clutter/background class is treated as a sparse background class and is therefore excluded from the reported statistics.

3) Implementation Details: Unless specified, we adopt ResNet-50 [3] and Swin Transformer [71] pretrained on ImageNet as the backbone network to ensure the detection performance of the simple architecture remains level. For all datasets, random flipping (0.75) is adopted as a data augmentation strategy. During the training process, AdamW [76] is adopted for optimization. The weight decay and base learning rate of AdamW are set as 0.05 and 0.0001. For stability training, we use warm-up (factor = 1/3, iterations = 1000) and multistep learning (training for 72 epochs, learning rate reduced to 1/10 at 45 and 65 epochs) for all datasets.

All experiments were implemented on a computer with 4× NVIDIA GeForce RTX 4090 (24 GB memory) and 4× NVIDIA A100 (40 GB memory).

B. Effectiveness Validation

We evaluate the effectiveness of the proposed method across three downstream tasks: horizontal object detection, oriented object detection, and semantic segmentation. By comparing our model with current SOTA methods across five RSI datasets, we demonstrate the consistent advantages of the proposed PMFM framework.

In our experimental setting, we adopted representative methods as baselines. For detection, we used RTMDet [72] (YOLO-style, one-stage, anchor-free), RetinaNet [56] (one-stage, anchor-based), S^2A-Net [5] (two-stage, anchor-based), and Swin Transformer [71] (Transformer-based). For segmentation, we used SegFormer [73] (Transformer-based), MaskFormer [74] and Mask2Former [75] (DETR-based).

1) Effectiveness on Oriented Object Detection: Beyond the lightweight baseline RetinaNet, we select two additional detectors, RTMDet and S^2A-Net, as benchmarks and verify the results on the DOTAv1.0 and DIOR-R datasets. The detailed experimental results are shown in Tables I and II, with each column presenting detailed data on the model's detection performance in a specific category. On DIOR-R, PMFM-FPN improves RetinaNet and RTMDet by 1.5% and 0.9% in mAP_{50}, respectively, with RTMDet reaching 67.8% mAP_{50}. Meanwhile, on the DOTAv1.0 dataset, S^2A-Net with PMFM-FPN achieves a modest improvement of 0.24% mAP_{50} over the baseline. When combined with the multiscale training strategy (MST), the performance is further improved to 79.92% mAP_{50}, corresponding to a gain of 0.50%. These results are consistent with the trends observed on other datasets and tasks, jointly indicating the complementary effect of PMFM in enhancing multiscale spatial representation during the feature fusion process.

2) Effectiveness on Horizontal Object Detection: We use Swin Transformer as the baseline detector and evaluate it on the SISP dataset. The detailed experimental results are reported

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TABLE II

COMPARATIVE EXPERIMENTS TO VALIDATE THE SOTA OF THE METHODS IN THIS ARTICLE ON THE DOTAV1.0 DATASET, WHERE "†" INDICATES THE REIMPLEMENTATION RESULTS. "∗" MEANS THE PROPOSED PMFM-FPN IS USED. "‡" MEANS THE MULTISCALE TRAINING IS USED

Methods	Venue	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP_{50}(%)
IENet [7]	arXiv'19	88.15	71.38	34.26	51.78	63.78	65.63	71.61	90.11	71.07	73.63	37.62	41.52	48.07	60.53	49.53	61.24
R^3Det [9]	AAAI'21	89.02	75.47	43.86	65.84	75.83	73.44	86.03	90.57	81.11	82.84	55.57	59.10	56.57	70.31	50.45	70.40
DHRec [82]	TPAMI'22	88.58	77.90	53.84	72.93	78.45	78.84	87.64	90.88	**88.78**	85.46	56.11	66.74	67.58	70.25	57.53	74.57
FCOSF [21]	TGRS'23	88.43	78.59	48.04	65.17	79.80	80.24	87.08	90.90	82.75	84.65	58.39	66.44	64.66	71.88	57.60	73.64
DCFL [83]	CVPR'23	89.22	79.15	50.95	71.29	77.44	73.92	84.22	90.90	84.07	84.73	57.07	66.53	65.84	69.51	55.09	73.33
DFDet [13]	TGRS'24	88.92	79.25	48.40	70.00	80.22	78.85	87.21	90.90	83.13	83.98	60.07	66.49	68.27	76.78	58.11	74.71
Xu [84]	GRSL'24	88.45	78.85	51.82	71.50	81.09	80.10	87.77	90.90	85.13	84.21	57.20	65.23	65.79	69.30	62.51	74.66
ARS-DETR [22]	TGRS'24	86.97	75.56	48.32	69.20	77.92	77.94	87.69	90.50	77.31	82.86	60.28	64.58	74.88	71.76	66.62	74.16
HRDet [12]	TCSVT'24	88.42	82.35	53.34	69.65	78.31	81.99	88.00	90.90	84.14	85.30	62.89	59.31	75.41	68.65	43.04	74.11
DCDet [85]	TGRS'24	**89.78**	83.46	57.23	79.62	80.12	83.99	88.41	90.90	86.16	87.22	66.84	**70.92**	77.29	78.67	67.87	79.23
OrientedFormer [20] ‡	TGRS'24	87.92	83.29	**58.92**	**80.90**	81.93	84.62	88.81	90.89	86.81	86.95	66.68	59.86	78.82	77.88	71.68	79.06
SOOD [81]‡	TGRS'24	88.30	**84.78**	57.88	80.78	79.97	84.19	88.74	90.84	85.95	87.56	**69.15**	69.13	77.32	79.22	**74.04**	79.86
FPNFormer [19]‡	TGRS'24	89.10	82.41	56.15	81.24	82.05	85.16	88.74	90.89	85.78	87.07	68.93	68.47	77.93	78.65	68.35	79.39
S^2A-Net [5] †	TGRS'22	87.95	75.16	51.00	70.58	78.66	84.23	88.27	90.86	84.16	84.34	55.41	61.55	73.18	73.44	59.16	74.53
S^2A-Net*(**ours**)		88.36	78.79	51.32	71.60	80.90	84.88	88.50	90.80	81.27	84.72	53.85	58.71	71.98	75.45	60.53	**74.77(+0.24)**
S^2A-Net [5] ‡	TGRS'22	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
S^2A-Net*(**ours**) ‡		89.05	82.73	57.48	80.65	**82.66**	**85.85**	**89.12**	90.88	87.26	**87.71**	67.56	67.43	**79.06**	**81.34**	70.03	**79.92(+0.50)**

TABLE III

EFFECTIVENESS VERIFICATION EXPERIMENTS IN THE HORIZONTAL OBJECT DETECTION TASK ON THE SISP DATASET. "*" MEANS THE PROPOSED PMFM-FPN IS USED

Methods	AP_{CS}	AP_{PL}	AP_{AS}	AP_{MS}	mAP_{50}(%)	$mAP_{50:95}$(%)
Faster R-CNN [86]	35.54	52.33	51.60	33.42	57.99	43.22
Mask R-CNN [87]	38.56	53.54	45.04	31.54	63.00	42.17
Cascade R-CNN [88]	45.38	56.24	**55.43**	36.83	64.40	48.47
Deformable-DETR [89]	21.16	36.35	12.25	0.33	34.77	17.52
RetinaNet [56]	31.67	47.01	47.57	22.7	60.39	37.24
HTC [90]	46.02	59.11	57.82	45.80	69.90	52.20
Swin Transformer [71]	43.67	53.20	54.97	48.27	69.69	50.03
Swin Transformer*(**ours**)	**46.31**	**57.08**	51.67	**58.81**	**72.38(+2.69)**	**53.47(+3.44)**

in Table III. Swin Transformer with PMFM-FPN achieves 72.38% mAP_{50} and 53.47% $mAP_{50:95}$, showing competitive performance among the compared detection methods. Specifically, Swin Transformer with the proposed module showed notable accuracy gains across multiple categories, particularly for PL (from 53.20% to 57.08%) and MS (from 48.27% to 58.81%).

These improvements benefit from PMFM, especially CaMPF, which progressively enhances the global semantic representations from shallow to deep layers, enabling better aggregation of features across varying object sizes and spatial complexities.

3) Effectiveness on Semantic Segmentation: We select three popular segmentation models, i.e., SegFormer [73], MaskFormer [74], and Mask2Former [75] as the baseline detectors and verify the results on the ISPRS Potsdam and Vaihingen datasets. As shown in Tables IV and V, the PMFM-FPN optimized benchmark segmentation models achieved impressive accuracy. On the ISPRS Potsdam dataset, the Mask2Former optimized with PMFM-FPN achieved competitive performance, reaching 85.89% mIoU and 92.27% mF1-scores. The second place was held by the PMFM-FPN-optimized SegFormer, which achieved 85.11% mIoU and 91.84% mF1-scores. On the ISPRS Vaihingen dataset, PMFM-FPN-optimized models also demonstrated commendable accuracy levels. Although they did not achieve SOTA performance, they still demonstrated favorable performance. Specifically, the PMFM-FPN-optimized MaskFormer achieved the highest IoU score of 86.14% for the Impervious Surface category, surpassing the second-place GAGNet-S (IoU: 85.80%) by 0.34%. Notably, GAGNet-S employs additional knowledge distillation techniques, whereas the MaskFormer features a simpler structure and relies solely on basic training strategies.

These results indicate the effectiveness of the PMFM, which embeds frequency–phase characteristics into positional encoding while simultaneously modeling spatial contexts and capturing critical high-frequency components (e.g., texture patterns, edge discontinuities). This frequency–spatial fusion mechanism demonstrates particular efficacy in semantic segmentation tasks requiring fine-grained texture discrimination.

C. Ablation Study

In this section, we conduct ablation studies to validate the effectiveness of individual components in the proposed PMFM. To align with the validation process of the baseline model FPNFormer, we selected RetinaNet as the benchmark detector for oriented object detection tasks.

1) Effectiveness of PhasePE: We first verify the effectiveness of the proposed PhasePE in Tables VI–VIII, which provide an intuitive comparison of the current mainstream position encodings (e.g., RelativePE, SinPE) under the same baseline detector, RetinaNet. On the DIOR-R dataset, PhasePE achieves higher mAP_{50} than the variants without positional encoding and those using RelativePE or SinPE. These results support the effectiveness of PhasePE in combining frequency-domain and spatial-domain modeling. In contrast, the baseline FPNFormer does not explicitly exploit frequency–phase information in positional encoding, which may limit its spatial representation capability. In addition, PhasePE introduces a RelativePE-style learnable weight matrix as an adaptive balancing unit to fuse frequency–phase information with standard relative-position priors, thereby improving spatial structure representation.

2) Effectiveness of CaMPF: Furthermore, we validate the impact of the proposed CaMPF, with results presented in Table VIII. On DIOR-R, FPNFormer with PhasePE obtains 60.1% mAP_{50}, while the full CaMPF design improves the performance to 61.1% mAP_{50}. This result is consistent with the lower right part of Fig. 2, demonstrating the focus-enhancing effect of CaMPF on the target, benefiting from the

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TABLE IV

EFFECTIVENESS VERIFICATION EXPERIMENTS WITH DIFFERENT MODELS. WE SELECTED POPULAR SEMANTIC SEGMENTATION MODELS AS THE BASELINE AND VALIDATED THEM ON THE ISPRS POTSDAM DATASET. THE HIGHLIGHTED TEXT INDICATES THE BEST SCORE. "*" MEANS THE PROPOSED PMFM-FPN IS USED. EVERY CLASS-WISE IOU IS SHOWN IN THE TABLE. "IMP. SURF." AND "LOW. VEG." INDICATE IMPERVIOUS SURFACE AND LOW VEGETATION

Methods	Venue	Imp. Surf.	Building	Low. Veg.	Tree	Car	mIoU(%)	mF1(%)
STransFuse [91]	JSTARS'21	83.93	90.44	74.72	75.63	86.61	82.27	90.48
ST-UNet [92]	TGRS'22	79.19	86.63	67.89	66.37	79.77	75.94	86.13
WSSS [93]	TGRS'23	85.91	91.57	75.35	76.82	88.86	83.70	90.98
EMRT [94]	TGRS'23	85.34	91.11	74.62	76.38	86.43	82.78	90.59
KFRNet [95]	GRSL'24	86.09	91.66	75.50	77.24	89.30	83.96	91.15
MSTNet-S [96]	TGRS'24	82.50	89.76	73.83	73.82	90.46	75.55	-
TCNet [97]	TGRS'24	**86.74**	**94.20**	76.75	76.66	90.63	85.00	91.73
LSENet [98]	GRSL'24	80.87	88.82	70.60	71.92	80.68	78.58	87.85
Lu [99]	TGRS'24	82.87	88.23	71.03	67.49	83.49	78.62	-
GAGNet-S [100]	TGRS'24	83.41	89.43	73.89	73.90	89.91	82.11	-
SegFormer [73]	NIPS'21	86.44	93.05	76.63	78.55	90.17	84.96	91.74
SegFormer*(**ours**)		86.52	92.71	**77.28**	78.95	90.09	**85.11(+0.15)**	**91.84(+0.10)**
Mask2Former [75]	CVPR'22	86.55	92.50	76.41	**79.75**	91.68	85.38	91.98
Mask2Former*(**ours**)		**87.54**	92.99	77.20	79.18	**92.54**	**85.89(+0.51)**	**92.27(+0.29)**

TABLE V

EFFECTIVENESS VERIFICATION EXPERIMENTS WITH DIFFERENT MODELS. WE SELECTED POPULAR SEMANTIC SEGMENTATION MODELS AS THE BASELINE AND VALIDATED THEM ON THE ISPRS VAIHINGEN DATASET. THE HIGHLIGHTED TEXT INDICATES THE BEST SCORE. "*" MEANS THE PROPOSED PMFM-FPN IS USED. EVERY CLASS-WISE IOU IS SHOWN IN THE TABLE

Methods	Venue	Imp. Surf.	Building	Low. Veg.	Tree	Car	mIoU(%)	mF1(%)
UNetformer [101]	ISPRS'22	85.13	92.58	66.66	83.29	**82.27**	81.98	89.85
ST-UNet [92]	TGRS'22	76.36	82.98	57.79	72.53	61.48	70.23	82.15
CMTFNet [102]	TGRS'23	84.67	92.07	68.07	**83.50**	82.17	**82.10**	**89.96**
LSENet [98]	GRSL'24	77.74	84.26	62.83	75.99	62.11	72.59	83.82
Lu [99]	TGRS'24	76.14	82.92	62.18	72.55	52.75	69.31	-
GAGNet-S [100]	TGRS'24	85.80	92.04	68.63	79.69	78.01	80.84	-
MSTNet-S [96]	TGRS'24	85.52	**93.29**	68.40	79.31	80.31	81.36	-
SegFormer [73]	NIPS'21	84.88	90.57	70.62	79.49	70.34	79.18	88.16
SegFormer*(**ours**)		84.71	90.02	71.01	79.73	71.83	**79.46(+0.28)**	**88.37(+0.21)**
MaskFormer [74]	NIPS'21	85.78	90.72	70.78	80.20	75.28	80.55	89.06
MaskFormer*(**ours**)		**86.14**	91.77	71.42	80.04	76.26	**81.13(+0.58)**	**89.41(+0.35)**
Mask2Former [75]	CVPR'22	85.78	91.59	**72.12**	80.70	73.02	80.64	89.10
Mask2Former*(**ours**)		85.64	91.70	71.31	80.18	75.54	**80.87(+0.23)**	89.04

TABLE VI

EFFECTIVENESS VERIFICATION EXPERIMENTS WITH DIFFERENT POSITION ENCODING METHODS. WE USE RETINANET WITH FPNFORMER AS THE BASELINE DETECTOR AND VALIDATE IT ON THE DIOR-R DATASET. "–" MEANS THE DETECTOR WITHOUT POSITION ENCODING METHODS

Methods	–	RelativePE [25]	SinPE [24]	PhasePE
mAP_{50}(%)	59.6	59.4	59.9	**60.1**

TABLE VII

PERFORMANCE COMPARISON OF SELECTED TYPICAL FREQUENCY-DOMAIN COMPONENTS COMBINED WITH PHASEPE METHOD: FULL SPECTRUM, AMPLITUDE, AND PHASE

PhasePE	Spectrum	Amplitude	Phase
mAP_{50}(%)	60.1	60.6	**61.1**

TABLE VIII

ABLATION EXPERIMENTS IN DIOR-R DATASETS. WE SELECT RETINANET AS THE BASELINE DETECTOR. "CA" MEANS CROSS-ATTENTION, AND "MF" MEANS MAMBA FUSION, WHICH TOGETHER FORM THE CAMPF MODULE. "✓" MEANS THE DETECTOR WITH PROPOSED MODULE

Neck	PhasePE	CA	MF	mAP_{50}(%)
FPNFormer [19]		✓		59.9
PMFM-FPN	✓	✓		60.1
PMFM-FPN	✓		✓	60.6
PMFM-FPN		✓	✓	61.0
PMFM-FPN	✓	✓	✓	**61.1**

progressive fusion process between cross-attention and Mamba Fusion. In our framework, CaMPF facilitates high-quality interaction between multiscale spatial and global features, thereby enhancing robustness and representation capability for RSI interpretation. The focusing effect of the proposed Mamba Fusion within CaMPF is visually substantiated in Fig. 6(c) and (d). Comparative visualization analysis reveals that this specialized component effectively amplifies feature responses of latent targets while suppressing irrelevant background noise.

3) Effectiveness of PhasePE-CaMPF: As shown in Table VIII, the results indicate that the primary performance gains originate from the frequency–spatial collaborative modeling paradigm jointly established by PhasePE and the progressive fusion mechanism in CaMPF, rather than from PhasePE or any single component alone. The comparison between CaMPF with and without PhasePE further

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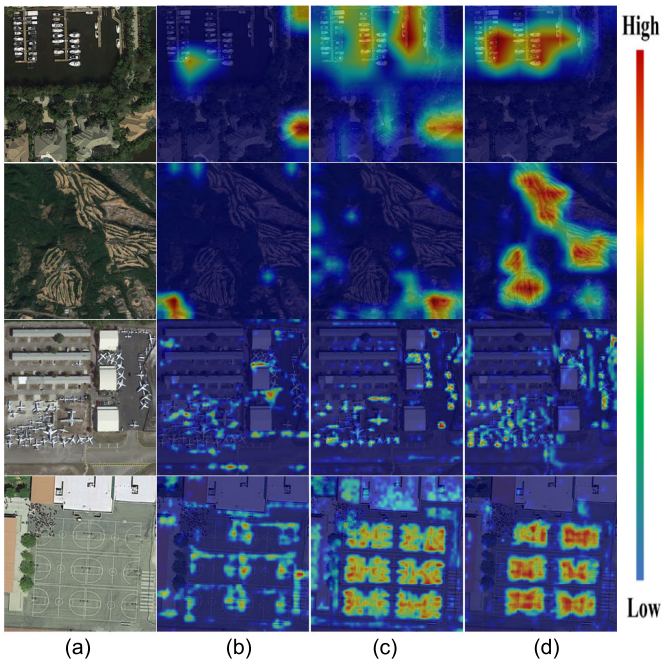


Fig. 6. Visualization of gradient-based feature maps during feature fusion. (a) Input. (b) FPN. (c) PMFM-FPN~CA, which denotes the variant without Mamba fusion in CaMPF, and (d) PMFM-FPN, which denotes the full version. The feature maps are extracted from different layers (down-top) $\{P2, P3, P4, P5\}$ of the FPN and PMFM-FPN networks based on RetinaNet.

TABLE IX

ABLATION EXPERIMENTS WITH DIFFERENT FPNS. WE SELECTED RETINANET AS THE BASELINE DETECTOR AND VALIDATE IT ON THE DIOR-R DATASET. "−W/−WO" MEANS THE MODULE WITH OR WITHOUT PMFM

Neck	FPN [37] (-w/-wo)	PAFPN [49] (-w/-wo)	FPNFormer [19] (-w/-wo)
mAP_{50}(%)	**54.6** / 48.4	**52.7** / 47.8	**61.1** / 59.9

demonstrates that PhasePE offers additional discriminative benefits, and its full potential is realized when integrated with CaMPF's progressive fusion process. The complementary roles of the two components are also consistent with the feature behavior observed in the model. Within CaMPF, cross-attention extracts spatial cues from PhasePE-enhanced features and interacts with adjacent levels to highlight informative regions, while Mamba Fusion applies state-space scanning to propagate and refine these responses across scales. This cooperative process strengthens target-focused activations and suppresses background interference, as illustrated in the visualization results in Fig. 6.

4) Effectiveness of PMFM: The proposed PMFM establishes a frequency–spatial collaborative modeling paradigm through PhasePE-CaMPF integration, achieving multiscale spatial feature representation for RSI comprehension. As quantified in Table IX, our plug-and-play PMFM delivers consistent enhancements across mainstream FPN variants: +6.2% (FPN), +4.9% (PAFPN), +1.2% (FPNformer) mAP_{50}. Additional improvements are observed on different detectors, such as +1.5% on RetinaNet (Table I) and +2.69% on Swin Transformer (Table III). Visualizations in Figs. 6 and 7 further demonstrate that PMFM strengthens target focus and suppresses background interference.

Extensive experiments have demonstrated the effectiveness of the method and PMFM can be integrated into various FPNs and detector architectures and is suitable for detection and segmentation tasks. Moreover, as described in Section IV-B and IV-C in experiments results, these results validate the model's versatility, confirming its capacity to generalize across different interpretation tasks, benchmark datasets, detectors, and FPNs.

5) Additional Verification: To further analyze the role of phase information, we compare the phase-based variant with other frequency-domain variants in Table VII. The results show that the phase variant achieves the highest mAP_{50} of 61.1%, compared with 60.6% for the amplitude-only variant and 60.1% for the full-spectrum variant. Although the numerical margin between phase and amplitude is relatively small, the phase variant shows a consistent advantage in this setting. From the perspective of Fourier analysis, amplitude mainly reflects intensity-related information, such as brightness, color, and global energy distribution, whereas phase is more closely associated with structural and geometric information, such as edges, contours, and spatial layout. Therefore, this observation suggests that phase-aware representations may provide useful complementary structural cues for complex RSIs.

In addition, we compare the performance of different component designs in the proposed method. The upper part of Table X compares different encoder choices in PhasePE. Compared with the classical ViT and the convolution-based ConvNeXt, the selected Swin Transformer achieves better overall performance. The lower part of Table X compares different fusion strategies in CaMPF. The proposed CaMPF achieves the best overall performance among the compared strategies, including lightweight convolutional fusion and the variant that reverses the order of the two stages. As described in Section III, CaMPF follows a "focus-then-refine" strategy, where cross-attention first emphasizes informative cross-scale spatial regions, and the subsequent Mamba-based fusion module propagates and refines these responses through long-range selective scanning. These results support the adopted design of CaMPF and are consistent with the feature behavior observed in Fig. 6.

D. Comparison With Related Methods

To further evaluate PMFM, we compare it with two groups of similar methods: Mamba-based methods and frequency-domain methods.

1) Mamba-Based Architectures: We compare PMFM with several recent Mamba-based methods, including LS-MambaNet [68], which employs large kernels for long-range modeling, OriMamba [69], which adopts a hybrid pyramid design for local–global integration, and DFLAN [70], which focuses on contextual modeling for small objects. As shown in Table XI, PMFM achieves competitive performance compared with these representative Mamba-based methods. Different from existing methods that mainly employ Mamba for contextual modeling or spatial scanning, PMFM is designed to further enhance object-aware spatial information modeling, thereby facilitating accurate localization in complex RSI scenes.

2) Frequency-Domain Processing Methods: We compare PMFM with several representative frequency-domain modeling methods. The corresponding results are reported in

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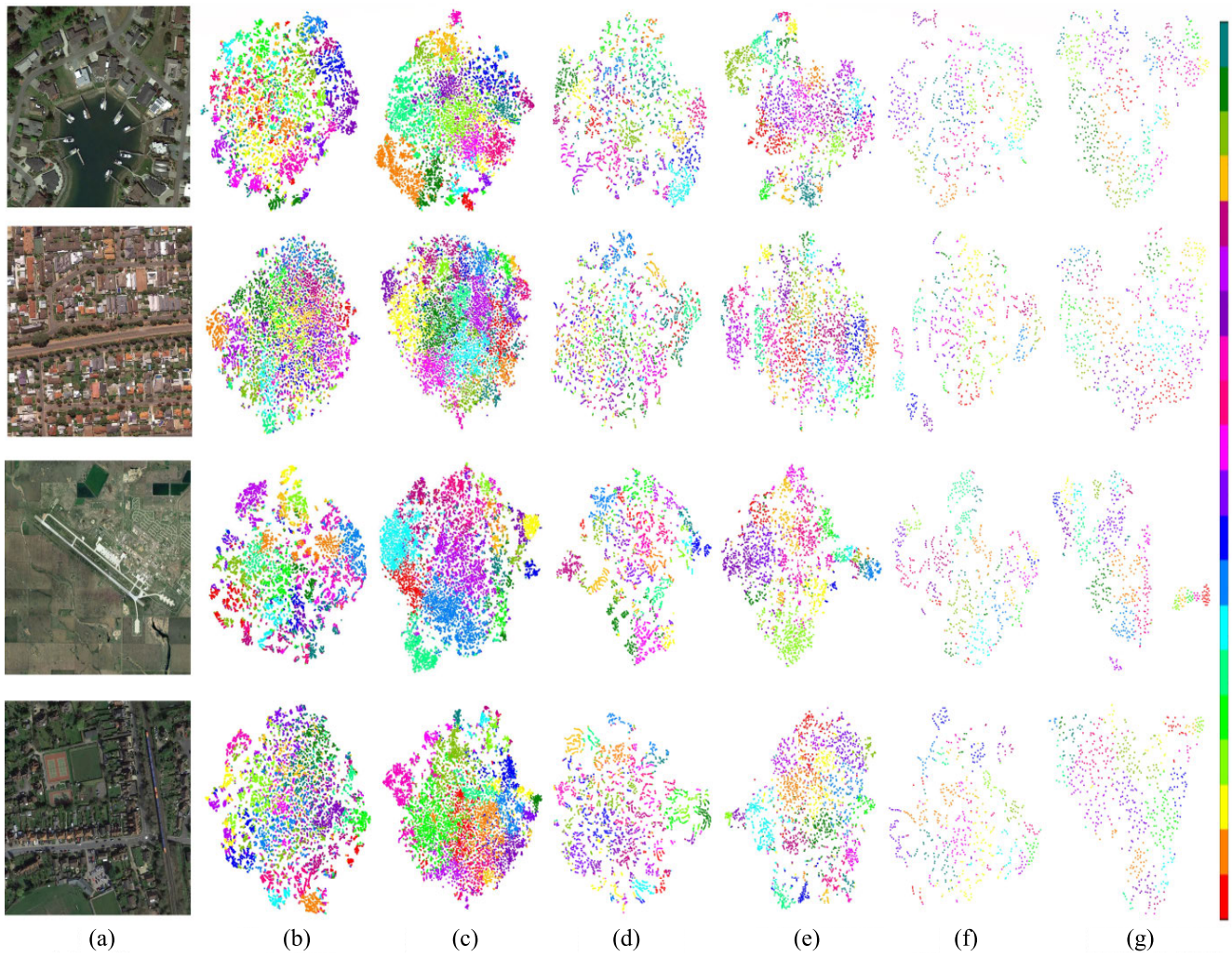


Fig. 7. t-SNE visualization comparison of FPN and PMFM-FPN at the P_3, P_4, P_5 layers. Each point represents a sample in the compressed feature space, where proximity reflects similarity in the high-dimensional space. The proposed PMFM-FPN demonstrates tighter clustering within each class and clearer separation across classes, especially in complex scenes. This suggests that PMFM strengthens the spatial information representation of the feature fusion process and improves target-specific classification capability from complex backgrounds. (a) Image. (b) FPN(P3). (c) PMFM-FPN(P3). (d) FPN(P4). (e) PMFM-FPN(P4). (f) FPN(P5). (g) PMFM-FPN(P5).

TABLE X

ABLATION STUDIES OF DIFFERENT ENCODERS IN PHASEPE AND FUSION STRATEGIES IN CAMPF ON THE DIOR-R DATASET, USING RETINANET AS THE DETECTION BASELINE. "L-M" DENOTES THE VARIANT THAT COMBINES A LIGHTWEIGHT CONVOLUTION WITH THE PROPOSED MAMBAFUSION. "M-C" DENOTES THE VARIANT THAT PERFORMS MAMBAFUSION BEFORE CROSS-ATTENTION, I.E., THE REVERSE ORDER OF THE PROPOSED DESIGN. "–W" MEANS THE MODULE IS USED

Methods	APL	APO	BF	BC	BR	CH	ESA	ETS	DAM	GF	GTF	HA	OP	SH	STA	ST	TC	TS	VH	WM	$mAP_{50}(\%)$
Ablation on different encoder designs in PhasePE																					
-w, ViT [4]	70.5	25.0	**76.7**	79.9	29.9	**72.7**	78.0	58.1	26.5	77.7	74.8	41.7	44.6	78.7	**74.8**	65.5	**80.8**	51.7	**43.3**	64.7	60.8
-w, ConvNeXt [103]	69.1	**31.1**	74.5	80.0	28.8	**72.7**	78.2	62.1	28.6	78.0	75.0	38.2	43.3	78.2	72.4	63.4	79.4	50.5	41.7	64.4	60.5
Ablation on different fusion strategies in CaMPF																					
-w, L-M	62.7	30.7	72.2	**80.2**	29.6	72.6	77.8	**63.8**	26.4	**78.5**	75.7	39.4	43.5	77.7	70.6	62.4	80.3	**52.8**	42.0	64.0	60.1
-w, M-C	**71.4**	29.2	73.6	79.2	30.5	**72.7**	**79.0**	62.9	26.8	78.4	73.2	38.7	**45.4**	78.3	72.9	63.9	80.0	52.1	41.6	**65.2**	60.7
PMFM (ours)	70.7	26.9	75.5	79.1	**31.9**	72.5	78.0	62.7	**29.2**	78.4	**77.4**	**42.3**	44.9	**79.0**	73.1	**66.3**	80.3	48.5	43.1	62.3	**61.1**

Tables XII and XIII. As shown in these tables, PMFM achieves competitive performance over existing frequency-domain methods in both object detection and semantic segmentation tasks. These results indicate that incorporating frequency-domain information into PMFM is beneficial for enhancing spatial representation in RSI interpretation tasks.

E. Generalization Experiments

1) Generalization Validation in RSI Interpretation: To further evaluate the generalization and effectiveness of the proposed method, we conduct experiments using various datasets and models. The detailed experimental results are presented in Tables I–V, respectively. By employing a block-based integration strategy, the PMFM-FPN optimized detector

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TABLE XI
PERFORMANCE COMPARISON WITH THE MAMBA-BASED METHODS ON THE DOTAV1.0 DATASET

Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP_{50}(%)
LS-MambaNet [68]	**90.02**	**86.88**	**62.46**	**84.04**	79.53	**86.27**	88.75	88.55	87.58	70.42	**72.66**	**78.74**	77.61	68.67	**76.12**	79.89
OriMamba [69]	88.80	84.14	56.49	74.73	81.47	85.77	88.50	90.44	86.82	85.57	60.79	69.83	77.96	69.50	72.63	78.23
DFLAN [70]	89.63	85.17	55.84	78.92	79.84	85.66	88.43	**90.88**	**87.94**	86.84	69.19	68.02	77.40	**81.89**	69.51	79.68
PMFM (ours)	89.05	82.73	57.48	80.65	**82.66**	85.85	**89.12**	**90.88**	87.26	**87.71**	67.56	67.43	**79.06**	81.34	70.03	**79.92**

TABLE XII
COMPARATIVE EXPERIMENTS WITH OTHER FREQUENCY-DOMAIN INFORMATION PROCESSING METHODS ON THE DOTAV1.0 DATASET (ORIENTED OBJECT DETECTION)

Detection	FDOL [104]	FSAR [105]	OII [106]	CoF-Net [107]	**PMFM (ours)**
mAP_{50}(%)	**80.41**	66.00	75.84	77.20	79.92

TABLE XIII
COMPARATIVE EXPERIMENTS WITH OTHER FREQUENCY-DOMAIN INFORMATION PROCESSING METHODS ON THE ISPRS POTSDAM DATASET (SEMANTIC SEGMENTATION)

Segmentation	FFGNet [108]	FDNet [109]	FAENet [110]	SSCNet [111]	**PMFM (ours)**
mIoU(%)	82.48	84.92	83.58	82.55	**85.89**

is easily achieved by aligning the channel computation structures of the feature extraction and prediction heads. As shown in Table III, the PMFM-FPN optimized Swin Transformer achieved improvements of 2.64% in mAP_{50} and 3.44% in $mAP_{50:95}$ scores compared to the baseline Swin Transformer model in the HBB object detection task. Similarly, RTMDet and S^2A-Net equipped with PMFM-FPN show consistent improvements in oriented object detection, further supporting the adaptability of the proposed feature fusion mechanism across different detection architectures. These advancements collectively enhance the model's generalization across diverse detection scenarios, boosting accuracy and robustness in both tasks.

To validate the cross-task adaptability, we extended PMFM-FPN to semantic segmentation. As evidenced in Tables IV and V, PMFM-FPN consistently enhanced performance across all benchmarks. On the ISPRS Potsdam dataset, PMFM-FPN brings consistent improvements to the baseline segmentation models in terms of both mIoU and mF1. Specifically, SegFormer and Mask2Former equipped with PMFM-FPN perform better than their corresponding baselines, suggesting that the proposed feature fusion mechanism is also beneficial for semantic segmentation. Specifically, SegFormer with PMFM-FPN showed notable accuracy gains in three out of five categories, such as impervious surface, low vegetation, and tree. Similarly, Mask2Former with PMFM-FPN improved accuracy in four out of five categories, including impervious surface, building, low vegetation, and car. This improvement benefits from PMFM-FPN's enhanced spatial information understanding, enabling better aggregation of similar classes at segmentation edges. In the ISPRS Vaihingen dataset, we supplemented MaskFormer, a high-precision baseline model, in pursuit of higher precision performance. As shown in Table V, PMFM-FPN consistently improves the baseline segmentation models on the ISPRS Vaihingen dataset. SegFormer, MaskFormer, and Mask2Former equipped with PMFM-FPN generally achieve higher mIoU scores than their corresponding baselines. In terms of mF1, SegFormer and MaskFormer also show performance improvements. These results suggest that PMFM-FPN can be effectively adapted to different tasks and model architectures, showing consistent performance improvements across the evaluated settings.

2) Auxiliary Validation in Low-Light Scenarios: To evaluate the behavior of phase-aware representations under degraded illumination conditions, we conduct additional experiments on the ExDark dataset. Although ExDark is a general low-light natural-image dataset rather than a remote sensing dataset, it provides an auxiliary visual benchmark for analyzing the robustness of phase-aware representations under challenging illumination conditions. As shown in Table XIV, the phase-based branch consistently achieves better performance than both the amplitude-based branch and the baseline. From a signal-processing perspective, amplitude mainly reflects intensity-related information and is thus more susceptible to illumination degradation, whereas phase preserves structural cues, such as contours and spatial layout, more stably. This observation suggests that phase-aware modeling can provide useful structural information under challenging low-light visual conditions.

F. Parameters and Computational Scale

We provide an analysis of the model's computational complexity, including FLOPs, parameters, and FPS, in Table XV. Compared with the SOTA Transformer-based fusion network, i.e., FPNFormer, PMFM-FPN introduces only a small increase in computational cost and parameter scale while maintaining inference speed comparable to that of the baseline models. Notably, benefiting from the high-optimization CUDA of the 2-D-FFT and the lower linear complexity of vision Mamba, the additional overhead introduced by our strategy is modest for modern detection stacks.

G. Visualization Analysis

To further analyze the working mechanism of PMFM and understand the reasons behind its performance improvement, we provide two types of feature visualization results, i.e., gradint-based feature maps and t-SNE. In addition, we compare detection results in complex RSI scenes to further verify the applicability and effectiveness of the proposed PMFM.

1) Gradient-Based Visualization: The effectiveness of PMFM can be clearly displayed in Fig. 6(d), where PMFM shows stronger target-focused activations and redced background responses compared with the baseline. Furthermore, we selected four types of typical scenarios (ship with harbor, golf field, plane, basketball court) to validate the facilitation efficacy of the proposed method, as shown in Fig. 6.

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TABLE XIV
COMPARATIVE EXPERIMENTS OF AMPLITUDE- AND PHASE-BASED VARIANTS OF THE PROPOSED PMFM ON THE EXDARK DATASET

Methods	Bicycle	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Motorbike	People	Table	$mAP_{50}(\%)$
RetinaNet [56]	**81.2**	**68.7**	**57.5**	83.0	72.4	76.8	61.1	58.3	83.3	**74.4**	**70.3**	49.5	69.7
PMFM (Amplitude)	64.9	43.3	31.7	74.4	55.8	54.5	34.9	37.0	57.3	53.2	53.0	28.3	49.0
PMFM (Phase)	80.5	67.5	57.1	**83.5**	**72.7**	**79.9**	**63.4**	**60.4**	**85.7**	**74.4**	69.4	**51.1**	**70.5**

TABLE XV
COMPARISON OF PARAMETER SCALE AND COMPUTATIONAL COMPLEXITY WITH OTHER FEATURE FUSION NETWORKS

Neck	FPN	PAFPN	FPNFormer	**PMFM-FPN**
FLOPS(G)	215.92	221.96	263.25	277.76
Parameters(M)	36.42	38.78	49.06	57.38
FPS	11.8	9.5	8.2	8.3

The results indicate that PMFM-FPN produces stronger target-focused responses while reducing background interference. Notably, in the scenes of a ship with a harbor, golf field, and basketball court, the complete version of PMFM-FPN demonstrates superior target perception capabilities compared to the PMFM-FPN$\widetilde{\text{CA}}$, without Mamba fusion), thanks to its global feature modeling process from shallow to deep layers. In the relatively spatially distributed ambiguous airplane scene, the PMFM-FPN shows a marked increase in attention to airplane targets, maintaining a focused highlight compared to the baseline model.

2) t-SNE Visualization: To better understand the facilitation of the proposed PMFM in the representation of spatial information, an unsupervised feature visualization method, t-SNE, was used to describe the feature clustering and separation in complex scenarios of RSIs. As shown in Fig. 7, the visualization was performed using multilevel features P_3, P_4, P_5 extracted from the DIOR-R dataset under oriented object detection. Each cluster represents semantically and spatially coherent groups of targets or background regions, demonstrating that the proposed PMFM-FPN achieves robust clustering within each class and clearer separation across classes in complex scenes. This suggests that PMFM strengthens the spatial information representation during the feature fusion process and improves target-specific classification capability in complex backgrounds of RSIs.

Furthermore, PMFM-FPN reduces feature distribution confusion caused by complex scenes and improves the separation of different feature representations. Specifically, the proposed PMFM-FPN achieves a more distinct sample separation compared to the baseline FPN in the P_3 layer, with feature points displaying a more unified distribution rather than overlapping or mixed clusters. This indicates that the issue of feature confusion between the interest targets and background noise in the early layers has been improved. In the P_4 and P_5 layers, PMFM-FPN continues to maintain clear separation and tighter clustering among feature points, demonstrating robust spatial information modeling capabilities. This improvement suggests that PMFM-FPN excels in complex scenes and in handling target orientation variations, exhibiting superior perceptual and predictive performance across challenging spatial distributions. These observations indicate that PMFM-FPN is better able to handle complex scenes and target orientation variations, which is consistent with the gradient-based feature maps.

3) Detection Visualization: As shown in Fig. 8, PMFM generates more accurate and stable object localization by strengthening multiscale spatial representations, particularly for small objects, densely distributed targets, and instances embedded in complex backgrounds. In contrast, the baseline detector frequently fails to separate adjacent objects and exhibits higher rates of false positives and missed detections. These visual comparisons further confirm the effectiveness of the proposed frequency–global modeling strategy in improving foreground–background separation and overall detection robustness.

V. DISCUSSION

A. Analysis of Phase Information

To better understand the potential role of phase information within the proposed framework, we analyze it from two complementary perspectives: state-space modeling and signal processing, based on the empirical observations and the data processing flow. From a state-space perspective, phase information is not introduced as a simple positional bias. Instead, the phase branch provides structural cues that are fused into the input representation before the SS2D selective scanning process. Since Mamba performs input-conditioned selective modeling, the fused phase-aware representation may affect the subsequent feature transition and selective representation process along the scanning path. From this perspective, phase information can be interpreted as a structural cue that may help support selective feature transitions along the scanning path. From a signal-processing perspective, phase information is closely related to structural patterns such as object contours and boundaries. When SS2D scanning traverses object-background transitions, these structural cues may provide useful information for enhancing the distinction between foreground structures and background regions. In this sense, phase-aware features may contribute to more structurally consistent long-range dependency modeling along the scanning path, instead of merely serving as an additional bias term.

B. Model Performance: Detection Versus Segmentation

The difference in performance across the two tasks primarily results from their distinct spatial reasoning requirements. Object detection relies heavily on region-level foreground–background discrimination, which is strongly aligned with the strengths of frequency–phase cues, as phase variations tend to highlight object contours, structural saliency, and foreground transitions. Semantic segmentation, on the other hand, depends on dense, pixel-level boundary refinement; these fine-grained details are less directly captured by phase-dominant representations. Consequently, the proposed

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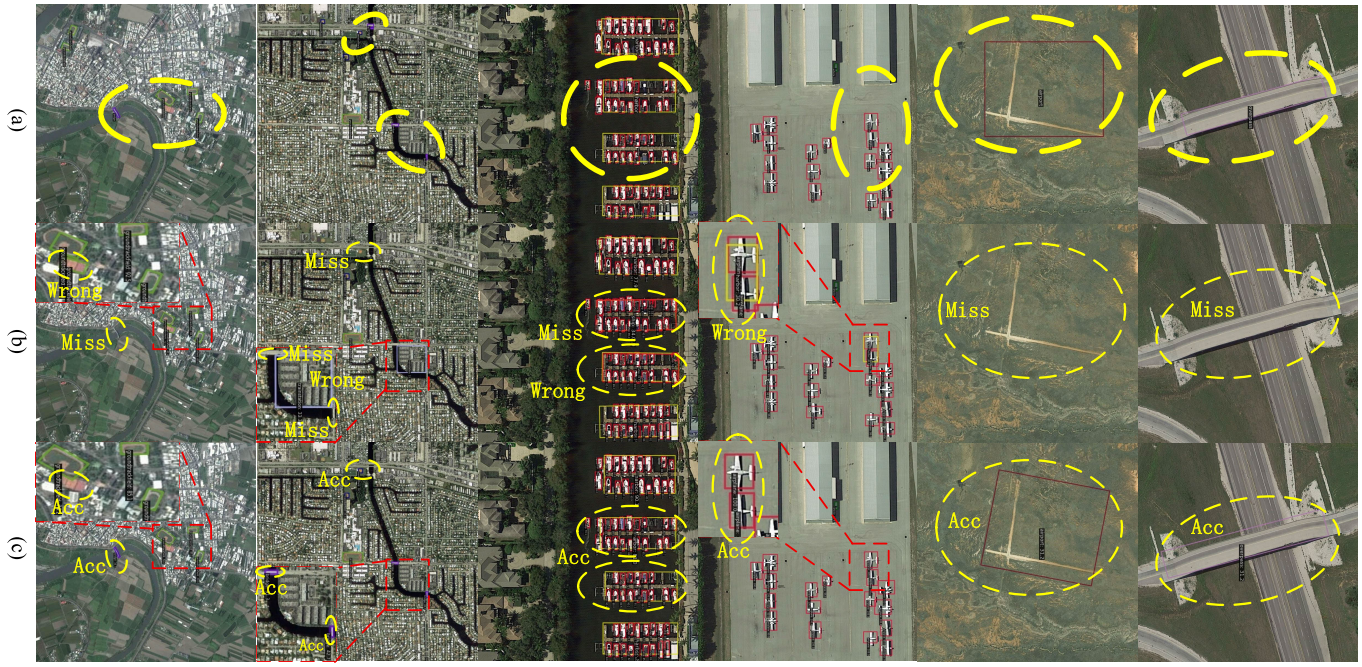


Fig. 8. Detection results on remote sensing images with multiscale objects. (a) Ground truth, (b) Baseline RetinaNet, and (c) PMFM-FPN. With the integration of PMFM, the detector substantially improves representation and localization of multiscale objects, enabling balanced attention to both large and small targets, and achieving superior accuracy and robustness under complex scenarios compared with the baseline.

frequency–phase enhancement yields more noticeable gains in detection tasks, where object-level localization benefits greatly from strengthened structural cues. Overall, the proposed phase-aware spatial representation achieves consistent improvements across multiple tasks. Although the magnitude of the gains varies among different tasks, this cross-task consistency supports the effectiveness of introducing frequency–phase information as a general auxiliary representation for remote sensing image interpretation.

VI. Conclusion

In this article, we presented the PMFM, a generic and flexible feature fusion framework for remote sensing image interpretation. By incorporating frequency–phase information into positional encoding and progressively modeling global–local dependencies through vision Mamba and cross-attention, PMFM enhances multiscale spatial representation in complex scenes. Furthermore, we developed a PMFM-FPN using a cross-scale concatenation strategy, demonstrating that PMFM can be effectively integrated into various detection and segmentation frameworks. Extensive experiments on five benchmark datasets across three remote sensing image interpretation tasks show that PMFM-FPN consistently improves baseline models for oriented object detection, horizontal object detection, and semantic segmentation, while achieving competitive performance compared with existing methods. These results demonstrate the effectiveness and generalization capability of the proposed method across diverse object scales, orientations, and scene complexities.

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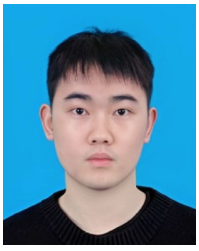
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